FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2021

PAPER –1

TEST DATE: 18-01-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
For each question, choose the option(s) corresponding to (all) the correct answer(s)
Answer to each question will be evaluated according to the following marking scheme:

Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
Negative Marks **: –2** In all other cases.

Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

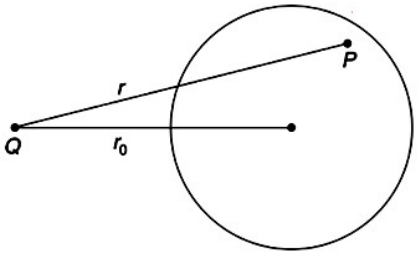
1. A trough has two identical inclined segments and a horizontal segment. A ball is released on the top of one inclined part and it oscillates inside the trough. Friction is large enough to prevent slipping of the ball. Time period of oscillation is T. Now the liner dimension of each part of the trough is enlarged four times. Then new time period will be

trough is enlarged four ti
period of oscillation of th



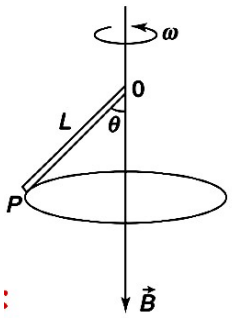
(A) T
(B) 2T
(C) 3T
(D) 4T

2. A conducting shell having no charge has radius R. A point charge Q is placed in front of it at a distance r_0 from its centre. Potential due to charge induced on the surface of the shell at a point P inside the shell is (Distance of point P from charge Q is r)



(A) $\frac{kQ}{r_0}$
(B) $\frac{kQ}{r}$
(C) $\frac{kQ}{r_0} - \frac{kQ}{r}$
(D) $\frac{kQ}{r} - \frac{kQ}{r_0}$

3. A conducting rod (OP) of length L rotates in form of a conical pendulum with an angular velocity ω about a fixed vertical axis passing through its end O. There is a uniform magnetic field B in vertically downward direction. The rod makes an angle θ with the direction of the magnetic field. The emf induced across the ends of the rod is



(A) $\frac{B\omega L^2 \sin^2\theta}{2}$
(B) $\frac{B\omega L^2 \cos^2\theta}{2}$
(C) $\frac{B\omega L^2 \tan^2\theta}{2}$
(D) $\frac{B\omega L^2}{2}$

4. Nuclei of radioactive element A are being produced at rate t^2 at time t. The element A has decay constant equal to λ. If N is the number of active nuclei of element A at any time t then $\frac{dN}{dt}$ was found to be minimum at $t = t_0$. N at time $t = t_0$ is

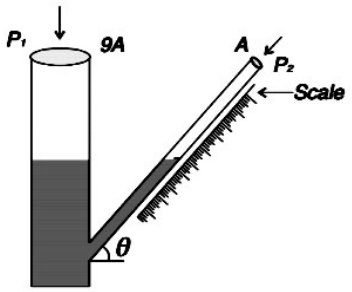
(A) $\frac{t_0 - xt_0^2}{x^2}$

(B) $\frac{t_0 - 2\lambda t^2}{\lambda^2}$

(C) $\frac{4t_0 - xt_0^2}{\lambda^2}$

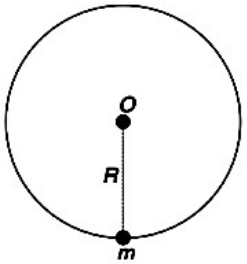
(D) $\frac{2t_0 - \lambda t_0^2}{\lambda^2}$

5. A manometer has a vertical of cross-sectional area 9A and an inclined arm having area of cross section A. The density of the manometer liquid has a specific gravity of 0.74. The scale attached to the inclined arm can read up to $\pm$ 0.5 mm. It is desired that the manometer shall record pressure difference ($P_1 - P_2$) up to an accuracy of $\pm$ 0.09 mm of water. To achieve this, the inclination angle θ of the inclined arm should be



(A) $\sin^{-1}(0.13)$
(B) $\sin^{-1}(0.03)$
(C) $\sin^{-1}(0.33)$
(D) $\sin^{-1}(0.23)$

6. A disc of mass M = 2m and radius R is pivoted at its centre. The disc is free to rotate in the vertical plane about its horizontal axis through its centre O. A particle of mass m is stuck on the periphery of the disc. The frequency of small oscillations of the system about its equilibrium position.



(A) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$

(B) $\frac{1}{2\pi}\sqrt{\frac{g}{2R}}$

(C) $\frac{1}{\pi}\sqrt{\frac{g}{2R}}$

(D) $\frac{1}{\pi}\sqrt{\frac{g}{R}}$

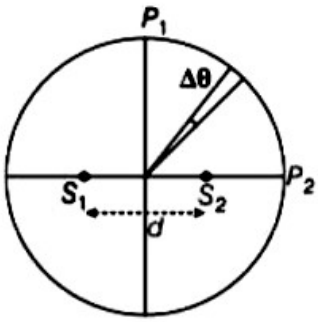
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. A student performed the experiment of determination of focal length of a concave mirror by u-v method using as optical bench of length 1.5 m. The focal length of the mirror used in 24 cm. The maximum error in the location of the image can be 0.2 cm. The sets of (u, v) values recorded by the student (in cm) are (42, 56), (48, 48), (60, 40), (66, 33), (78, 39). The data set(s) that cannot come from experiment and is (are) incorrectly recorded, is (are)

(A) (42, 56)
(B) (48, 48)
(C) (66, 33)
(D) (78, 39)

8. Two coherent monochromatic point source S_1 and S_2 of wavelength λ = 600 nm are placed symmetrically on either side of the centre of the circle as shown in the figure. The sources are separated by a distanced d = 1.8 mm. This arrangement produces interference fringes visible as alternate bright and dark spots on the circumference of the circle.



The angular separation between two consecutive bright spots is $\Delta\theta$. Which of the following options is/are correct?

(A) The angular separation between two consecutive bright spots decreases as we move from P_1 to P_2 along the first quadrant

(B) A dark spot will be formed at the point P_2

(C) The total number of fringes produced between P_1 and P_2 in the first quadrant is close to 3000

(D) At P_2 the order of the fringe will be maximum

9. The polonium isotope $^{210}_{84}Po$ is unstable. It emits an α-particle with energy of 5.30 MeV. The atomic mass of $^{210}_{84}Po$ is 209, 9829u and atomic mass of $^{4}_{2}He$ is 4.0026 u, then choose the correct options.

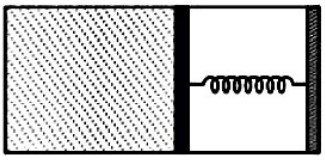
(A) Daughter nucleus is $^{206}_{82}Po$

(B) Disintegration energy is 5.30 MeV

(C) Disintegration energy is 5.40 MeV

(D) Mass of daughter nucleus is 205.9745 u

10. An ideal monoatomic gas is confined in a horizontal cylinder by a spring-loaded piston (as shown in the figure). Initially, the gas is at temperature T_1, pressure p_1 and volume v_1 and the spring is in its relaxed state. The gas is the heated very slowly to temperature T_2, pressure p_2 and volume v_2. During this process, the piston moves out by a distance x. Ignoring the friction between the piston and the cylinder, then choose the correct statements.



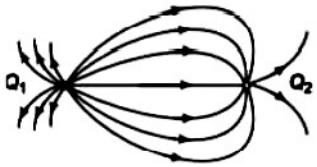
(A) If $V_2 = 2V_1$ and $T_2 = 3T_1$, then the energy stored in the spring is $1/4(p_1V_1)$.

(B) If $V_2 = 2V_1$ and $T_2 = 3T_1$, then the change in interval energy is $3p_1V_1$

(C) If $V_2 = 3V_1$ and $T_2 = 4T_1$, then the work done by the gas is $\frac{7}{3}p_1v_1$

(D) If $V_2 = 3V_1$ and $T_2 = 4T_1$, then the heat supplied to the gas is $\frac{1}{4}(p_1v_1)$

11. A few electric field lines for a system of two charges Q_1 and Q_2 fixed at two different points on the X-axis are shown in the figure. These lines suggest that



(A) $|Q_1| > |Q_2|$

(B) $|Q_1| < |Q_2|$

(C) at a finite distance to the left of Q_1, the electric field is zero

(D) at a finite distance to the right of Q_2, the electric field is zero

12. Specific heats of a gas varies with temperature. Now, choose the correct options, let hydrogen gas sample is taken in observation.

(A) At temperature around 60 K, $C_v = \frac{3}{2}R$

(B) About 250 K, $C_v = \frac{5}{2}R$

(C) About 750 K, $C_v = \frac{5}{2}R$

(D) About 4000K, $C_v = \frac{7}{2}R$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

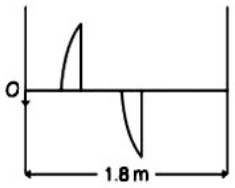
13. A binary star consists of two starts A (mass 2.2 M_S) and B (mass 11 M_S), where M_S is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is

14. A hydrogen atom is in n = 5 state makes a transition to it's ground state. Assuming, H-atom is initially at rest, find the recoil speed of H-atom.

15. A 5 m long cylindrical steel wire with radius 2×10^{-3} m is suspended vertically from a rigid support and carries a bob of mass 100 kg at the other end. Calculate the energy stored in the wire upto two decimal places. (For the steel wire: Young's modulus = 2.1×10^{11} Pa, density = 7860 kg-m^3.

16. Degree of freedom for a gas is 6. This gas does 25 J of piston work at constant pressure. Net heat absorbed by the gas during this process is …… .

17. A thin plano-convex lens of focal length f is split into two halves. One of the halves is shifted along the optical axis. The separation between object and image planes is 1.8 m. The magnification of the image formed by one of the half lens is 2. Find the separation between the halves.



18. A 100 kg gun fires a ball of 1 kg horizontally from a cliff of height 500 m. It falls on the ground at a distance of 400 m from the bottom of the cliff. If recoil velocity of gun is v, then find the value of v. (Take, g = 10 ms^{-2})

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. Twenty molecules of SO_3 will weigh as much as ____ molecules of oxygen.
(A) 100
(B) 50
(C) 15
(D) 8

20. An ideal gas expands according to the law $P^2V = \text{constant}$. on expansion, the temperature
(A) decreases
(B) increases
(C) remains constant
(D) none of these

21. The enthalpy of neutralization of a strong acid by a strong base is -57.32kJ mol^{-1}. The enthalpy of formation of water is -285.84 kJ mol^{-1}. The enthalpy of formation of equeous hydroxyl ion is
(A) + 228.52 kJ/mol
(B) -114.26 kJ/mol
(C) -228.52 kJ/mol
(D) +114.2 kJ/mol

22. Liquid ammonia ionizes to slight extent.
At -50°C, its self-ionization constant, $K = [NH_4^+][NH_2^-] = 10^{-30}M^2$. How many amide ions are present per ml of pure liquid ammonia? ($N_A = 6 \times 10^{23}$)
(A) 10^{-15}
(B) 10^{-18}
(C) 6×10^5
(D) 6×10^8

23. In the equation: $4M + 8CN^- + 2H_2O + O_2 \rightarrow 4[M(CN)_2]^- + 4OH^-$
Identify the metal M:
(A) Iron
(B) Copper
(C) Zinc
(D) Gold

24. Which of the following complex species does not involve d^2sp^3 hybridisation?
(A) $[CoF_6]^{3-}$
(B) $[Co(NH_3)_6]^{3+}$
(C) $[Fe(CN)_6]^{3-}$
(D) $[Cr(NH_3)_6]^{3+}$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

25. Which of the following are correct methods for the preparation of propanoic acid?

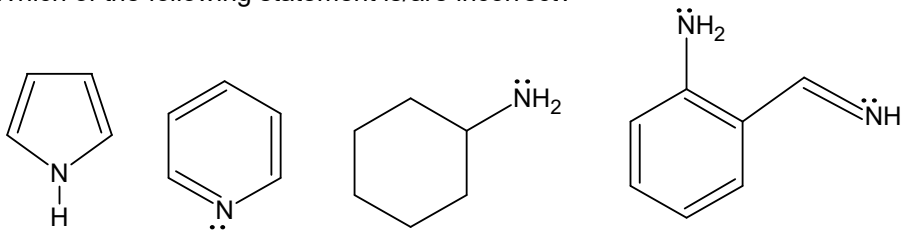
(A) $H_3C-CH=CH_2 \xrightarrow{HBr} \xrightarrow[Ether]{Mg} \xrightarrow[H_3^{\oplus}O]{CO_2}$

(B) $H_3C-C\equiv CH \xrightarrow[H_2O_2,\ ^{-}OH]{BH_3.THF} \xrightarrow{KMnO_4}$

(C) $H_2C=CH_2 \xrightarrow{HBr} \xrightarrow[Ether]{Mg} \xrightarrow[H_3^{+}O]{CO_2}$

(D) $H_3C=CH-\underset{\displaystyle CH_3}{\underset{|}{C}}-CH_3 \xrightarrow[Ag_2O]{O_3}$

26. Which of the following statement is/are incorrect?



(I) (II) (III) (IV)

(A) I and II are aromatic and have equal basic strength
(B) I is aromatic and II is anti-aromatic but II is stronger base than I
(C) The order of basicity of the above of the above compounds is IV > III > II > I
(D) III is non-aromatic

27. Which statement is/are correct in following?
(A) Glucose does not give 2,4-DNP test.
(B) Cane sugar give non-equimolar mixture of D-(+)-glucose and D-(-)-fructose.
(C) Cellulose does not give reaction with Tollens' Reagent.
(D) Fructose is reducing sugar.

28. Which of the following oxy anion(s) contains(s) P-O bond order equal to 1.5?
(A) $H_2P_2O_6^{2-}$
(B) $H_2PO_3^{-}$
(C) $H_2PO_4^{-}$
(D) $H_2PO_2^{-}$

29. Which of the following pairs of cations cannot be separate by using an NaOH solution?
(A) Fe^{3+}, Al^{3+}
(B) Cr^{3+}, Al^{3+}
(C) Sn^{2+}, Pb^{2+}
(D) Cu^{2+}, Pb^{2+}

30. Mercury is a liquid at 0°C because of:
(A) very high ionisation energy
(B) weak metallic bonds
(C) high heat of hydration
(D) high heat of sublimation

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. The edge length of the unit of LiCl (NaCl-like structure) is 520pm. Assuming that the lithium ion is small enough that the chloride ions are in contact, the ionic radius of chloride ion (in pm) is $\left(\frac{1}{\sqrt{2}} = 0.7\right)$.

32. An aqueous solution contains 12% (w/w) $MgSO_4$ and 9.5% (w/w) $MgCl_2$. If the sulphate dissociates up to 80%and chloride dissociates up to 60%, the boiling point of solution (in Kelvin) is (K_b of water = 0.785 K –kg/mol)

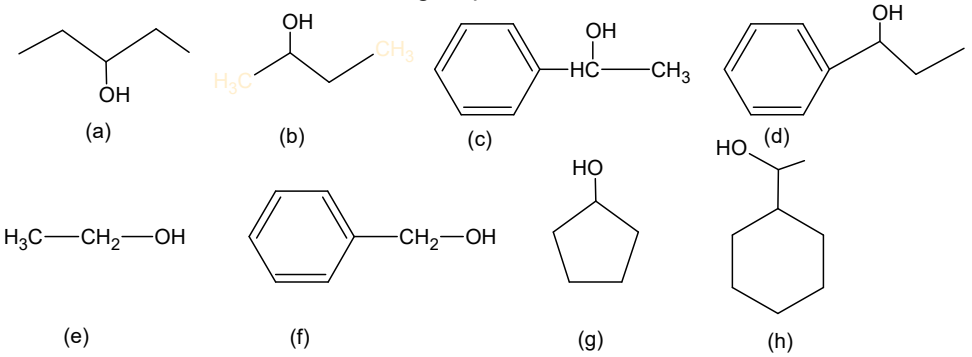
33. The reaction 2A + B $\rightarrow$ C + D goes to completion and follows the rate law $-\frac{d[A]}{dt} = K[A]^2[B]$. The sum of values of x and y in the following data:

Set	$[A_0]\times10^6$M	$[B_0]\times10^6$M	Half-life(s)
1	300	40	62.5
2.	300	60	x
3.	5	300	625
4.	10	300	y

34. A 5.0g sample of optically pure (+)-2-bromooctane was dissolved in 40 ml of CCl_4. The observed rotation was +2.7°, measured in a 10cm polarimeter tube.
What is the specific rotation of (+2)-2-bromooctane?

35. What is the percentage of tertiary chloride in products of the following reaction?
$CH_3CH(CH_3)CH_3$
Assume relative rates of abstraction of primary, secondary and tertiary hydrogen to be 1:4:6.

36. Find out number of alcohols that can give positive iodoform test.



Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. Value of $\sum_{r=1}^{10} {}^{15}C_r \cdot {}^{35}C_{10-r}$ is

(A) ${}^{10}C_{10}$

(B) ${}^{50}C_{10} - {}^{35}C_{10}$

(C) ${}^{40}C_{10}$

(D) ${}^{40}C_{10} - 1$

38. Let $f(x) = \int x^{\sin x}\left(1 + x\cos x \ln x + \sin x\right)dx$

And $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$, Find value of $\cos\left(f\left(\pi\right)\right)$?

(A) $\frac{\pi}{2}$

(B) π

(C) 0

(D) 1

39. Find unit vector which is orthogonal to vector $3\hat{i} + 2\hat{j} + 6\hat{k}$ and is coplanar with vectors $2\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} - \hat{j} + \hat{k}$ is

(A) $\frac{2\hat{i} - 6\hat{j} + \hat{k}}{\sqrt{41}}$

(B) $\frac{2\hat{i} - 3\hat{j}}{\sqrt{13}}$

(C) $\frac{3\hat{j} - \hat{k}}{10}$

(D) $\frac{4\hat{i} + 3\hat{j} - 3\hat{k}}{\sqrt{34}}$

40. If $S_n = \sum_{r=1}^{n} \frac{2r+1}{r^2 + 2r^3 + r^2}$, then S_{10} equals

(A) 1

(B) $\frac{121}{120}$

(C) $\frac{120}{121}$

(D) $\frac{99}{100}$

41. If $x \in [-1, 0)$, then $\cos^{-1}\left(2x^2-1\right)-2\sin^{-1}x$ equals

(A) $-\frac{\pi}{2}$

(B) π

(C) $\frac{3\pi}{2}$

(D) -2π

42. If A, B, C are real numbers, then find value of $\begin{vmatrix} 1 & \cos(B-A) & \cos(C-A) \\ \cos(A-B) & 1 & \cos(C-B) \\ \cos(A-C) & \cos(B-C) & 1 \end{vmatrix}$

(A) $\cos A+\cos B+\cos C$

(B) $\cos A.\cos B.\cos C$

(C) independent of A, B, C

(D) $\cos A.\cos C+\cos B\cos C+\cos A\cos B$

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. If all roots of equation $z^4+az^3+bz^2+cz+d=0$ (a, b, c, d $\in$ R) are of unit modulus, then

(A) $|a|\le 4$

(B) $|b|\le 4$

(C) $|c|\le 4$

(D) $|b|\le 6$

44. Consider function $f(x)=\left(\frac{ax+1}{bx+2}\right)^x$ where $a^2+b^2\neq 0$, then $\lim\limits_{x\to\infty} f(x)$ is

(A) exists for all values of a and b

(B) zero for 0 < a < b

(C) non-existent for a > b > 0

(D) $e^{\frac{-1}{a}}$ or $e^{\frac{-1}{b}}$ for $a=b$

45. If the length of tangents from A, B, C to the incircle of ΔABC are 4, 6, 8 then which of the following is (are) correct ?

(A) Area of ΔABC is $12\sqrt{6}$

(B) Ex-radii of ΔABC are in HP

(C) Sides of ΔABC are in AP

(D) radius of incircle of ΔABC is $\frac{4\sqrt{6}}{3}$

46. If $f(x)=\left(\sin^2 x-1\right)^n$, $n\in N$, then $x=\frac{\pi}{2}$ is a point of

(A) Local maximum if n is odd
(B) Local minimum if n is odd
(C) Local maximum if n is even
(D) Local minimum if n is even

47. Let f(x) be a non-negative continuous and bounded function for all $x \geq 0$. If $\cos x f'(x) \leq (\sin x-\cos x) f(x) \forall x \geq 0$ then which of the following is/are correct ?

(A) $f(6)+f(5)>0$

(B) $x^2-3x+2+f(7)=0$ has two distinct solution

(C) $f(5).f(7)-f(6)(5)=0$

(D) $\lim\limits_{x\to 6}\frac{f(x)-\sin(\pi x)}{(x-6)}=1$

48. There are 8 different lines, 7 different circles and 6 different equilateral triangles, then maximum number of points of intersection of

(A) lines only is 28
(B) circles only is 42
(C) triangles only is 90
(D) circles and triangles any is 384

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. If $x^2+2y^2-2xy-2\geq k(x+2y)\forall x, y\in R$, then find the number of integer values of k ?

50. If solution of differential equation $x^2.\frac{d^2y}{dx^2}+2x.\frac{dy}{dx}=12y$ is $y=Ax^m+Bx^{-n}$ then find maximum value of |m+ n|

51. Find area of region by two branches of $(y-x)^2=x^3$ and straight line x = 1 ?

52. Find value of $\int_0^1 \frac{\tan^{-1}x}{\cot^{-1}(1-x+x^2)}dx$?

53. TP and TQ are any two tangents to a parabola and the tangent at a third point R cuts them is P' and Q', then find $\frac{TP'}{TP}+\frac{TQ'}{TQ}$.

54. Rectangle ABCD has area 200 unit^2. An ellipse with area 200π unit^2 passes through A and C and has foci at B and D. find perimeter of rectangle ABCD ?